

Question number	Answer	Additional guidance	Mark
4(a)	the {length of DNA / sequence of (DNA) bases} that code for {amino acids / (poly)peptide / protein} (1)	ACCEPT nucleotides for bases primary structure of protein	(1)

Question number	Answer	Additional guidance	Mark
4(b)	An explanation that includes the following points: - because the mutation is in the gene coding for the CFTR (protein) (1) - therefore the CFTR (protein) does not function correctly (1) - credit details of dysfunction (resulting in very thick sticky mucus) (1) - therefore the mucus will be (very) {thick / sticky} (1)	ACCEPT mutation in CFTR gene ACCEPT change in structure e.g. reduced transport of chloride ions out of the cell sodium ions move into the cell water leaves the mucus and enters the cell	(3)

Question number	Answer	Additional guidance	Mark
4(c)	An answer that includes three of the following points: • couples (both) carrying one copy of the mutation can be identified (1) • they can then make (an informed) {decision / choice} (about having a child) (1) • credit an example of their options (1) • resulting in fewer babies being born who are homozygous (1)	ACCEPT couples who are (both) heterozygous / have a CF allele DO NOT ACCEPT choose which embryos to implant e.g not having a child / adoption / IVF DO NOT ACCEPT have an abortion ACCEPT two copies of the mutation fewer heterozygous babies born	(3)